

Total No. of Questions : 8]

SEAT No. :

PD4225

[Total No. of Pages : 2

[6403]-19

T.E. (Civil Engg.)

SOLID WASTE MANAGEMENT

(2019 Pattern) (Semester - VI) (301015 F) (Elective - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q.No. 1 or Q.No. 2, Q.No. 3 or Q.No. 4, Q.No. 5 or Q.No. 6, Q.No. 7 or Q.No. 8.*
- 2) *Figures to the right indicate full marks.*
- 3) *Draw neat figures wherever necessary.*
- 4) *Assume suitable data if necessary.*
- 5) *Use of scientific calculator is allowed.*

- Q1)** a) State the factors considered during evaluation of on-site processing equipments of solid waste. **[6]**
- b) Explain with sketch the Bangalore Method of Vermicomposting. **[6]**
- c) State the terms related to Solid waste management. **[6]**
- i) Prevention
 - ii) Minimisation
 - iii) Recycling
 - iv) Reuse

OR

- Q2)** a) State different methods of collection of Solid waste and explain any one with sketch. **[6]**
- b) List the factors affecting composting process of solid waste. **[6]**
- c) Write a note on material recovery facility. **[6]**
- Q3)** a) What are the stages of Anaerobic digestion. **[6]**
- b) Write the environmental impacts of waste to energy system. **[6]**
- c) What are the different types of incinerators? Explain any one. **[5]**

OR

P.T.O.

Q4) a) Describe the incineration technologies and air emissions and its control in detail. Explain the following: [6]

i) Pyrolysis

ii) Refuse-derived fuel

b) How does energy recovery relate to renewable energy. [5]

c) Write advantages and disadvantages of Plasma Gasification. [6]

Q5) a) Explain trench method of landfilling and state the factors to be considered for site selection. [6]

b) Write principles of Bioreactor landfill. [6]

c) Draw and explain with neat labeled sketch of single liner system of leachate control. [6]

OR

Q6) a) Write short notes on legal aspects of solid waste disposal. [6]

b) Describe the factors which affect production of leachate and landfill gas in the landfill? [6]

c) State any two advantages and disadvantages of Landfill method of solid waste disposal. [6]

Q7) a) Write objectives and major provision in construction and demolition (C&D) waste management rules - 2016. [6]

b) Suggest various management technologies for biomedical waste. [6]

c) Suggest various management technologies for slaughter waste. [5]

OR

Q8) a) Explain any one case study of material recovered from e-waste. [6]

b) Write objectives and major provision in plastic waste management rules - 2016. [6]

c) Explain EPA - identification of hazardous and toxic waste. Write down the characteristics of hazardous waste. [5]

